

Ecological Conservation Redlines

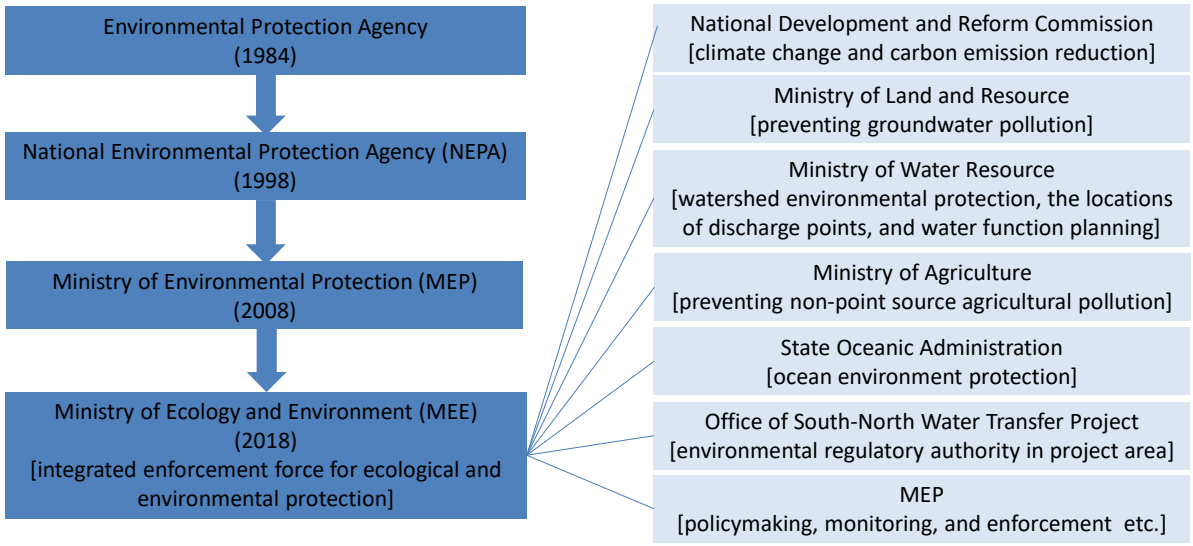
Dr. Tingjia Chen
School of Economics and Management
Harbin Institute of Technology (Shenzhen)

Learning Objectives

After taking this class, the students will be able to:

- *understand* what ecological redline is
- *acknowledge* how the ecological redlines are delineated
- *tell* the differences between ecological redlines and natural protection zone

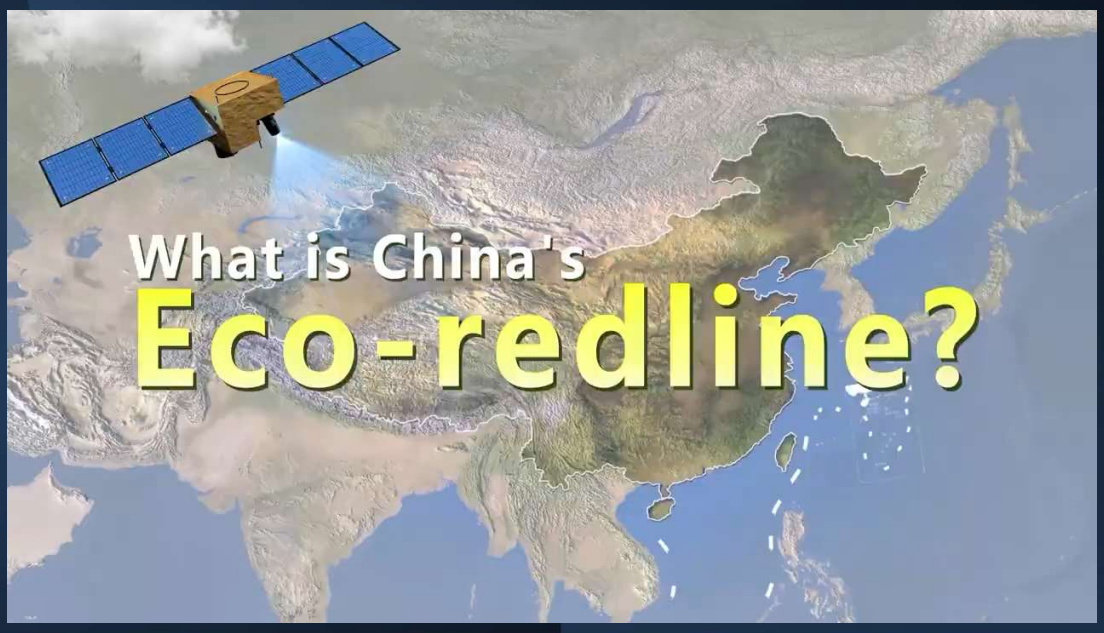
Background: the Evolution of Chinese National Environmental Agencies



Ecological Conservation Redlines

The Redlines

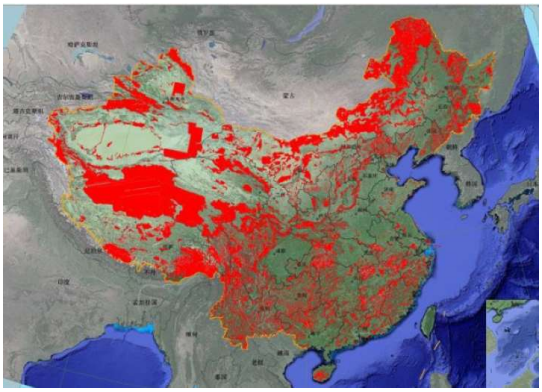
- Water Resources (by 2030)
  - Water use in total: total volume of water use will be limited to within 700 Bm^3
  - Water quality: the up-to-standard rate of the quality of water at water function zones will reach 95 % or higher
  - Water use efficiency: the volume of water use for generating every 10,000 yuan in industrial value added will decrease to 40 m^3 or lower
- Arable land: remains at or above the redline of 1.8 billion mu
- Forest land (by 2050): Tree covers remain or above 26% of total land use; 4.68 billion mu



What is ecological redline?

Ecological Redline I

- Ecological baseline area needs to provide ecosystem services to guarantee and maintain ecological safety, living environment safety, and biological safety.
- ***minimum*** ecological area needed to guarantee and maintain ecological safety and functionality, and biological diversity for national security.
- The naming of ecological protection red lines adopts the format "natural geographical unit + dominant ecological function + ecological protection red line", such as "Yanshan Water Conservation Ecological Protection Red Line".



What is ecological redline?

Ecological Redline II

- These areas form the baseline and lifeline of ecological security. They typically include regions vital for functions such as water conservation, biodiversity maintenance, soil conservation, windbreak and sand fixation, and coastal ecological stability.
- They also encompass environmentally sensitive and fragile areas affected by soil erosion, land desertification, rocky desertification, and salinization.



The Milestone of Ecological Conservation Redlines Policy

- The concept of ecological conservation redline was originally raised in 2011 by the State Council
- The MEP spent more than 6 months in discussing how to delineate ecological conservation redlines in 2012.
- Four pilot provinces have been identified by the end of 2012. (Inner Mongolia, Jiangxi, Hubei, Guangxi)
- The ecological redlines were written into the Environmental Protection Law of 2015.
- In 2017, the eco-redline policy started to be implemented nationwide.
- The “Ecological Redline Implementing Regulation” was drafted for public comments in October 2017.

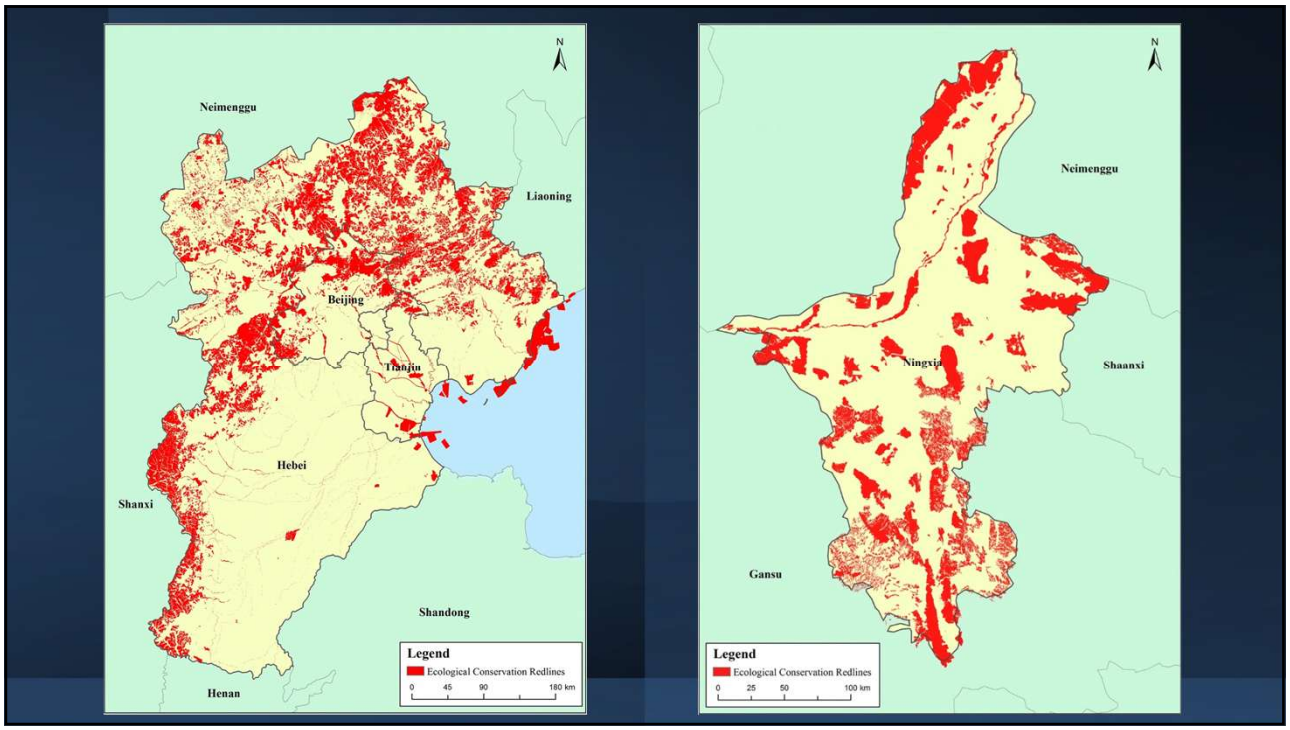
How are the ecological redlines delineated?

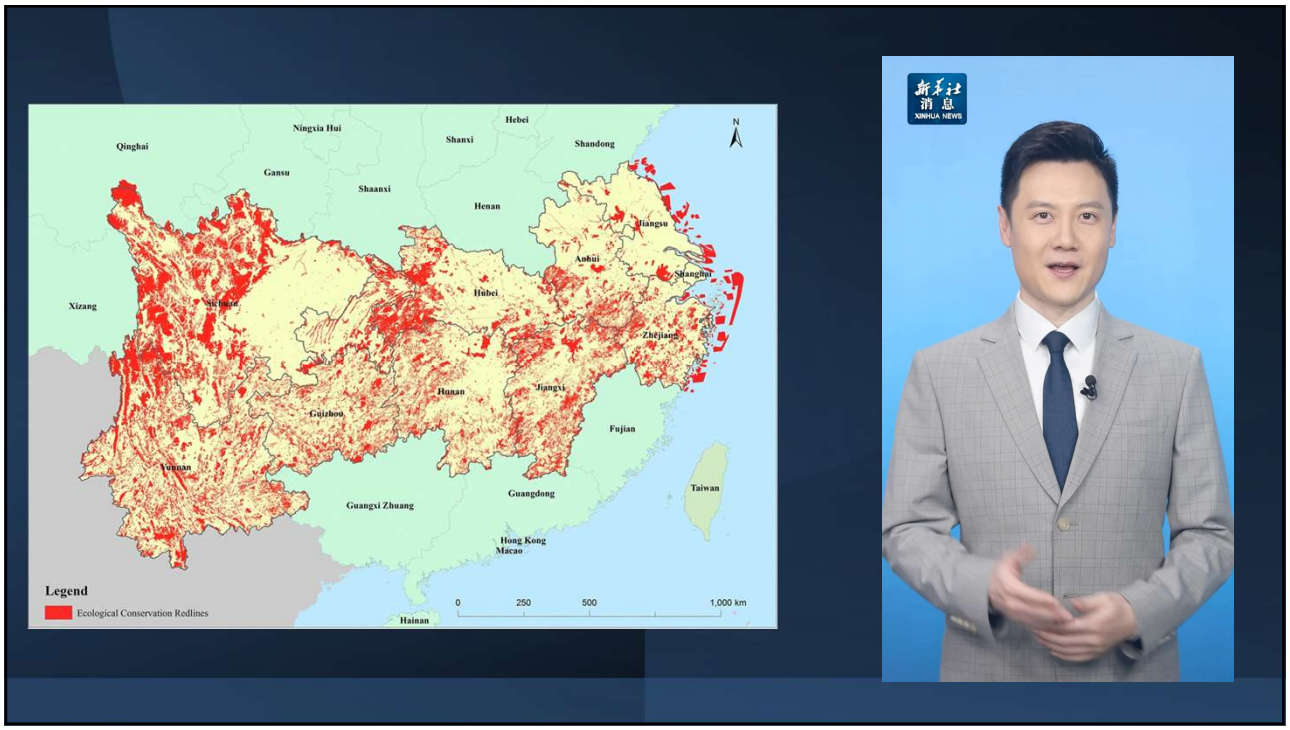
How are the ecological redlines delineated?

- Identify the critical areas for ecological conservation
- Delineate the boundaries of ecological redlines
- On-site investigation, determine boundary, and erect sign accordingly
- Properly deal with other land use areas within the redlines (Most provinces delineate 20%-30% of the land as ecological redline areas.)

Which areas are likely to be included in the ECR Area?

- Landscapes that *deliver valuable ecosystem services*, such as water conservation, soil conservation, wind breaks, and biodiversity conservation;
- *Ecologically sensitive areas*, such as areas of land sandification, water and soil loss, stony desertification, and salinization;
- *Legally established protected ecological areas*, such as nature reserves, national parks, forest parks, and wetland parks;
- Other areas that need protection, such as habitats of endangered species, natural coastlines, and plateaus with permafrost.


N
Neimenggu
Liaoning
Beijing
Tianjin
Hebei
Shanxi
Shandong
Henan
Legend
Ecological Conservation Redlines
0 45 90 180 km
N
Neimenggu
Ningxia
Shaanxi
Gansu
Legend
Ecological Conservation Redlines
0 25 50 100 km


N
Ningxia Hui
Hebei
Qinghai
Shanxi
Shandong
Gansu
Shaanxi
Henan
Jiangsu
Anhui
Shanghai
Xizang
Sichuan
Hubei
Zhejiang
Hunan
Jiangxi
Guizhou
Fujian
Yunnan
Taiwan
Guangdong
Guangxi Zhuang
Hong Kong
Macao
Hainan
Legend
Ecological Conservation Redlines
0 250 500 1,000 km
新华社
消息
XINHUA NEWS

How are the ecological redlines delineated?

Ecological “Redlines” in the Urban Area: the case of Shenzhen

- Shenzhen is the first metropolitan in China making policies regarding ecological *control* lines.
  - Shenzhen Basic Ecological Control Line Management Regulations (2005)
  - Implementation Opinions on Enforcing the “Shenzhen Basic Ecological Control Line Management Regulations” (2007)
  - Implementation Opinions on Further Standardizing the Management of the Basic Ecological Control Line (2013)
- It intends to deal with the tensions among population, land, resources, and environment. It is basically to prevent disordered city sprawl, thus delineating a basic line for ecological protection and future city development.
- Total area of Shenzhen is 1997.47 km^2; the land area within the limits of ecological control lines is about 974.5 km^2, which is about 48.76% of the city area.





Year	Permanent Population (10000 persons)	Permanent Registered Population (10000 persons)	Population Density	Land Area (sq.km)	Construction land Use (sq.km)
2010	1037.2	251.03	5208	1997.47	930
2020	1763.38	514.1	8828	1997.47	976
2035 (expected)	1900	1105	9512	1997.47	1105

However, ecological control line is different from ecological redline from several perspectives.

- Total area within the limits of ecological conservation redlines in Shenzhen is 401.59 km^2, which is about 20.11% of the land area of the city.
- Ecological control line is usually delineated by the local governments, while the ecological redlines policy is a more centralized policy with corresponding stricter accountability mechanisms.

How do ecological redlines being managed/regulated?

- The management of ecological protection red lines is, in principle, handled according to the requirements for areas where development is prohibited.
- All types of development activities that do not conform to the main ecological function are strictly forbidden, to ensure that **the ecological functions are not diminished, the area is not reduced, and the nature does not change**.
- Adjustments needed for the construction of national major infrastructure or significant projects for the protection of people's livelihoods should be organized and demonstrated by the provincial government, which should propose an adjustment plan. After the Ministry of Environmental Protection and the National Development and Reform Commission, in conjunction with relevant departments, have reviewed and given their opinions, the plan should be submitted to the State Council for approval.